

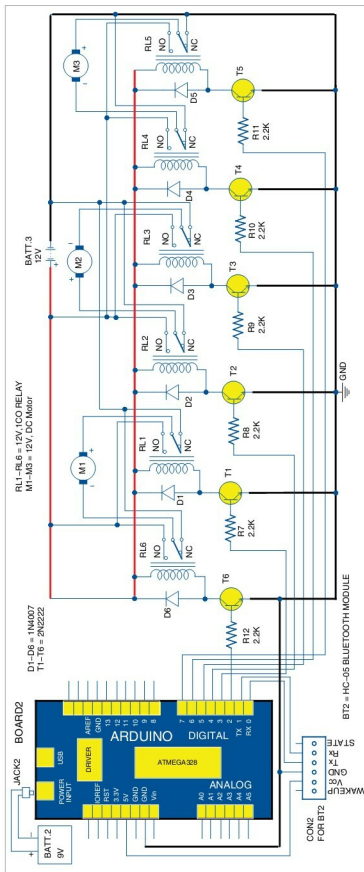


Fig. 4: Circuit diagram of receiver unit

(USB-to-UART) and also a terminal software. The author used a Tera Term software, which can be downloaded freely and easily from the Internet.

You need the following wiring connections between HC-05 module and CP2102 converter:

HC-05 (BT1)	CP2102
Vcc	+ 5V
GND	GND
Tx	Rx
Rx	Tx
Key	3.3V

After the connection, plug-in CP2102 module to USB port of your PC. Open Tera Term software from your PC and proceed as follows:

Step 1. Select 'Serial' option, followed by the COM port on which the CP2102 converter is connected

Step 2. Select Setup → Serial Port → Baudrate → 38400

The default baud rate for HC-05 is 38400 for command mode

Step 3. Under the setup menu, select terminal as 'CR + LF' for Transmit. Also tick the local echo. Thus you can see whatever you have typed or entered in the window

Step 4. Type 'AT + RMAAD' to remove all the previously connected devices.

Set the password same as for slave by typing the following commands in the edit window of Tera Term:

AT + PSWD = *****

AT + ROLE = 1. Here '1' is for master mode

AT + CMODE = 1 allows the module to connect to any address

AT + INIT. Type this command to initiate serial port. If you find error 17, don't worry, it means you have already started SPP

AT + INQ. Type this to find devices near master module. After a while, you will get a list of the devices. To pair with the slave module, type:

AT + LINK = "the address with commas"

Here, AT + LINK = 98d3,31,305f7b

Master module (BT1) is configured now.

Switch on both the BT1 and BT2 modules and see the status LEDs on both slave and master modules. Both modules should automatically connect to each other. If both LEDs start blinking twice in a second simultaneously, both devices are connected to each other and will remember each other even